

The SecOps Transformation and Your SOC

What Is SecOps?

SecOps is a collaboration between IT security and operations teams around cybersecurity. Like DevOps, it introduces shared responsibility between teams, removing communication barriers and streamlining cooperation. SecOps can help organizations streamline communication between teams, increase visibility of security issues, and identify and mitigate threats more efficiently, addressing the challenges of the traditional security operations center (SOC).

In many organizations, SecOps is evolving into [DevSecOps](#) - a broader collaboration between development teams, security, and IT operations.

What Challenges Does SecOps Address?

Security concerns and threats are rapidly evolving. New security vulnerabilities and new attack techniques are discovered on a daily basis. Attackers continue to develop new tools and technologies, and a traditional security operations center (SOC) based on Security Information and Event Management (SIEM) cannot keep up. Another key issue is the security skills shortage, which makes it difficult for the SOC to hire and retain security talent.

The SecOps challenges posed by traditional SOC environments include alert fatigue, complex investigations, lack of visibility in the IT environment, and lack of automation and orchestration. Security teams need to work hard to make sense of complex IT systems, without much cooperation from the teams responsible for building and maintaining them.

To make matters worse, when security incidents are discovered, security teams make recommendations but find it difficult to get those implemented. IT teams have their own priorities and schedules, and security concerns often take a back seat.

SecOps can address these concerns by making IT and SOC experts jointly responsible for security, and helping them work together to create an IT environment that can be easily monitored, and where it is easy to remediate security issues as soon as they are discovered.

How Will SecOps Change Your Organization?

The goal of SecOps is to improve cybersecurity by making security considerations a shared responsibility of IT and security teams. This involves several organizational changes:

- Overcoming the silo approach in traditional security management to join IT and security teams into one.
- Raising awareness of the impact of security considerations on business operations, and making every member of the team aware of and responsible for security.
- Giving team members the knowledge, tools, and organizational processes they need to work together on security matters across all areas of operations.
- Automating, simplifying, and standardizing security operations.
- Ensuring one toolset can be used by IT and security engineers, with consistent APIs that enable integration with other organizational systems.

Integrating SecOps into the SOC

Many organizations have a dedicated Security Operations Center (SOC), a team of security professionals who collaborate to identify and mitigate risks and defend against security threats.

SecOps refers to a set of tools, processes, and practices that a SOC may use to protect an organization and implement its security strategy. However, traditional SOC workflows are not compatible with a SecOps culture. Historically, the SOC was isolated from other teams within an organization—security staff performed their duties without interacting with anyone else.

Today's business culture requires closer collaboration across business areas. Decision-makers are increasingly treating security as a shared responsibility, embracing the concept of a modern SOC that encourages communication and collaboration between the security and operations teams.

There are several ways the SOC can start integrating its processes with the development and IT departments:

- **Distributing the SOC**—dismantle the security silo and spread SOC responsibilities across multiple departments, emphasizing incorporating security into the operations team's workflows. This approach enables SecOps and DevSecOps.
- **Creating a security center of excellence (COE)**—build a combined team that integrates the SOC with specific members from operations and development teams. Ensure everyone understands and implements security best practices.
- **Fostering a collaborative workplace culture**—open up the SOC so all staff members with some security impact can interact and work closely with security experts. There should

be an easy way for developers and engineers to consult with the security team, including senior members, on various issues.

Best Practices for SecOps Implementation

Conduct Red and Blue Team Training Exercises

SecOps team members can improve their security skills by participating in red and blue team exercises. The red team attacks the system using social engineering, port scanning, and vulnerability scans. In contrast, the blue team protects the system using vulnerability detection, security policy and tool evaluation, and analytics. The two teams train against each other and provide reports to help strengthen their overall SecOps capabilities.

Divide Processes Into Manual and Automated Workflows

Automation is essential for SecOps in large, fast-paced environments, allowing teams to monitor systems, detect anomalies, and identify vulnerabilities faster. Some security threats are possible to address automatically. However, some processes require manual involvement—for instance, complex incident response procedures and the creation of playbooks.

An effective SecOps strategy requires identifying which processes are suited to automation and what must remain manual. Striking a balance between manual and automated processes helps improve the SecOps team's ability to implement a fast, thorough response.

Implement SecOps Throughout the Delivery Pipeline

SecOps teams must implement their processes to prevent and mitigate threats at every stage of the software delivery pipeline. Traditionally, security teams focus on the production environment, waiting until the end of the development pipeline to start testing and scanning. SecOps pushes these processes to the beginning of the pipeline—for example, by performing vulnerability immediately when writing code. Throughout and after deployment, teams should perform various security checks and monitoring.

Prioritize Alerts

A SecOps team typically receives constant barrages of alerts that become increasingly difficult to manage. Prioritizing and filtering the most important alerts helps reduce the noise and allows the team to focus on urgent and high-risk events. Prioritization is the key to optimizing response effectiveness and managing the SecOps team's resources.

The SOC leadership can use various approaches to prioritize alerts, including data-driven (i.e., DLP), threat-driven, and asset-driven tools and strategies.

SecOps with HackerOne

HackerOne blends our SaaS platform with the world’s largest community of ethical hackers to find application vulnerabilities before they are exploited by bad actors. By catching these flaws before they become security events, fewer alerts will need to be sent to SecOps teams so they can focus on remediating incidents already in their queue. Once critical vulnerabilities are found, HackerOne Triage services can deduplicate reported issues and provide prioritized remediation guidance, enabling SecOps teams to increase their effectiveness.

[Learn more about the HackerOne platform](#)

Company

Leadership

Careers

Partners

Newsroom

Contact Us

Knowledge

Center

Application Security

Penetration Testing

Cloud Security

Hacking

Cybersecurity

Attacks

DevSecOps

Resources

Blog

Documentation

Leaderboard

Partner Portal

Resources

Contacted
by a  hacker?

Trust